

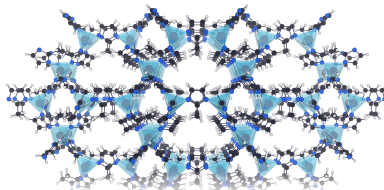


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Magnetism and Open-Shell Systems - Tutorial

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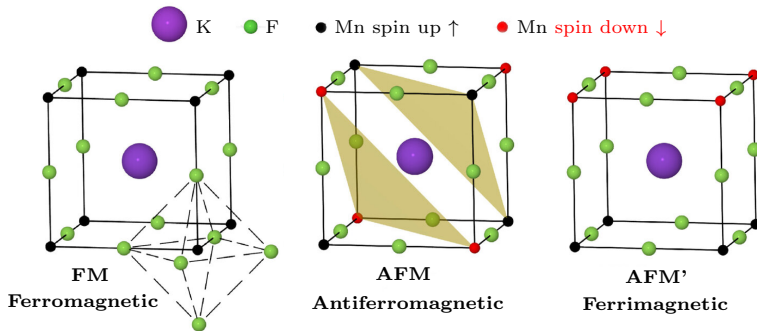
Theoretical Chemistry Group
University of Torino



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TORINO 8-12 SEP 2025

Example. A simple magnetic compound: KMnF_3

- Cubic crystalline system ($\text{Pm}\bar{3}\text{m}$ space group)
- One formula unit per unit cell and three atoms in the asymmetric unit
- Each Mn ion is at the center of a regular octahedron
- Fully ionic crystal with Mulliken atomic charges very close to the formal oxidation number ($q_{\text{K}} = +1|e|$, $q_{\text{F}} = -0.9|e|$ and $q_{\text{Mn}} = +1.8|e|$)
- Three different magnetic phases

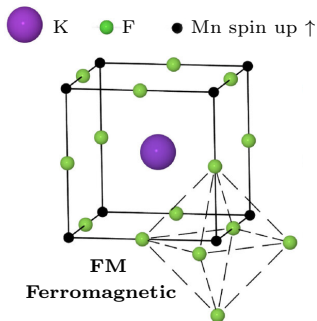


Which are the observables of interest?

First of all, the **relative stability of the different magnetic phases**.

In this tutorial, the FM and the AFM phases will be considered and analyzed.

The ferromagnetic phase of KMnF_3 crystal



Simplified version of the Heisenberg model :

$$\mathcal{H} = -J \sum_{i=1}^m S_i^z S_{i+1}^z$$

J is the magnetic exchange coupling constant

m are the number of lattice sites

S_i^z is the spin component along the z direction

Electronic configuration $\rightarrow$ K : $[\text{Ar}] 4s^1$ Mn : $[\text{Ar}] 3d^5 4s^2$ F : $[\text{He}] 2s^2 2p^5$

Oxidation state $\rightarrow$ K^{1+} : $[\text{Ar}]$ Mn^{2+} : $[\text{Ar}] 3d^5$ F^{1-} : $[\text{He}] 2s^2 2p^6$

The ferromagnetic solution is obtained by requiring 5 unpaired electrons per unit cell.

Since the unit cell contains only one transition metal atom, this requirement implies that the total spin projection in the unit cell along the z direction is

$$S_z^{\text{cell}} = 5/2$$

The ferromagnetic phase of KMnF_3 crystal - SPINLOCK keyword

The CRYSTAL directive is **SPINLOCK**, which requires the specification of two integer arguments:

1. first integer : value of $n_\alpha - n_\beta$
where n_α and n_β denote respectively the number of α and β spin electrons per cell
(in the case of ferromagnetic KMnF_3 the requirement is $n_\alpha - n_\beta = 5$)
2. second integer : number of self-consistent field (SCF) cycles with $n_\alpha - n_\beta$ fixed

Note : The SPINLOCK directive does not constrain unpaired electrons to localize at a certain nucleus. It simply states that the difference between the number of α and β spin electrons per cell needs to be equal to the fixed value $n_\alpha - n_\beta \rightarrow S_z^{\text{cell}} = (n_\alpha - n_\beta)/2$

[in the third block : single particle Hamiltonian and SCF control block]

SPINLOCK	Keyword to lock the spin difference $n_\alpha - n_\beta$
5 30	$n_\alpha - n_\beta$ and number of SCF cycles with $n_\alpha - n_\beta$ fixed
END	End of the third block input section

In the example above, the states are populated in such a way that $n_\alpha - n_\beta = 5$ for 30 SCF cycles and the constraint is then removed starting from the 31-st cycle on.

The five unpaired electrons are not constrained to be localized at the Mn ions sites. The SPINLOCK directive only constraints the total spin projection in the unit cell along the z direction to be equal to $5/2$.

The ferromagnetic phase of KMnF_3 crystal - SPINLOCK keyword

Exercise 1. SPINLOCK keyword to study the ferromagnetic phase of KMnF_3 crystal.

- Run CRYSTAL with file $\text{KMnF}_3.\text{d12}$ as an input file and analyze the output
- How can the spin density distribution be visualized?

The ferromagnetic phase of KMnF_3 crystal - SPINLOCK keyword

Exercise 1. SPINLOCK keyword to study the ferromagnetic phase of KMnF_3 crystal.

- Run CRYSTAL with file $\text{KMnF}_3.\text{d12}$ as an input file and analyze the output
- How can the spin density distribution be visualized?

Solution.

Spin density maps are certainly the best representations of spin density distribution.

However, useful information is gained from **Mulliken population analysis** (MULPOPAN directive in the SCF input section of CRYSTAL), that allows to see **how spin density is distributed among atoms and atomic orbitals**.

In this case, Mulliken population analysis shows very clearly that all the 5 unpaired electrons are almost completely localized on the Mn ion in the unit cell and populate the *d* atomic orbitals

→ this picture is in substantial agreement with the very simple model of an isolated ion in a crystal field, and indicates that the *d* atomic orbitals are hardly part of the valence structure of the crystal.

The ferromagnetic phase of KMnF_3 crystal - SPINLOCK keyword

Exercise 2. SPINLOCK keyword to study the ferromagnetic phase of KMnF_3 crystal.

Using the input file `KMnF3.d12` as a starting point, answer to the following question.

- What happens if the keyword SPINLOCK is not used?

The ferromagnetic phase of KMnF_3 crystal - SPINLOCK keyword

Exercise 2. SPINLOCK keyword to study the ferromagnetic phase of KMnF_3 crystal.

Using the input file `KMnF3.d12` as a starting point, answer to the following question.

- What happens if the keyword SPINLOCK is not used?

Solution.

If the SPINLOCK directive is omitted, there is **no guarantee a priori that the self-consistent cycle may proceed to the ferromagnetic solution spontaneously**, when starting from a guess Fock (Kohn-Sham) matrix of isolated atoms or ions.

In this case, if the SCF process is not driven by the SPINLOCK directive to the proper ferromagnetic phase spin configuration, it drifts towards an unstable metallic nonmagnetic state (i.e. a non polarized closed shell or diamagnetic phase).

The ferromagnetic phase of KMnF_3 crystal - SPINLOCK keyword

Exercise 3. SPINLOCK keyword to study the ferromagnetic phase of KMnF_3 crystal.

Using the input file `KMnF3.d12` as a starting point, answer to the following question.

- What happens if the keyword SPINLOCK is applied only for few SCF cycles?

Repeat the calculation by restricting the SPINLOCK keyword to only one SCF cycle.

The ferromagnetic phase of KMnF_3 crystal - SPINLOCK keyword

Exercise 3. SPINLOCK keyword to study the ferromagnetic phase of KMnF_3 crystal.

Using the input file $\text{KMnF}_3.\text{d12}$ as a starting point, answer to the following question.

- What happens if the keyword SPINLOCK is applied only for few SCF cycles?

Repeat the calculation by restricting the SPINLOCK keyword to only one SCF cycle.

Solution.

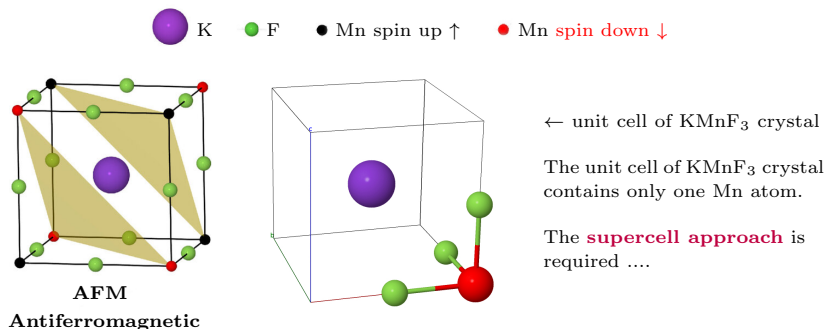
In general, the SPINLOCK directive **needs to be active until the self-consistent process has found a numerically stable path along which the system will converge to the required configuration** regularly.

Actually, in this case, the ferromagnetic solution can be obtained by applying the SPINLOCK constraint only once, at the beginning of the SCF cycle.

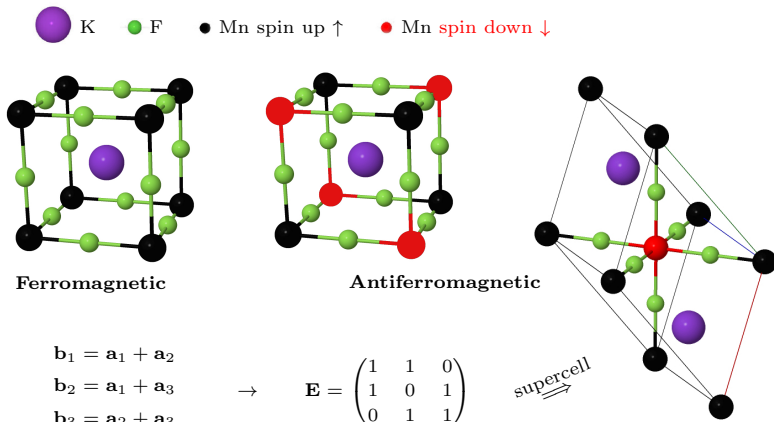
The antiferromagnetic phase of KMnF_3 crystal

Although, in principle, an infinite number of antiferromagnetic cells with $S_z^{\text{cell}} = 0$ can be tried to find the most stable phase, fortunately, it is well known that the **experimental antiferromagnetic phase** of KMnF_3 corresponds to that geometry in which **each Mn ion is surrounded by six nearest neighbouring Mn ions with opposite spin**.

To achieve this configuration, a **cell containing two nearest neighbouring Mn ions** has to be considered, and the symmetry relationship between them has to be broken, so that they may carry opposite spin, i.e. transform the primitive cubic lattice into a commensurate face-centered cubic super-lattice, which results from interpenetrating α and β -spin Mn sublattices.



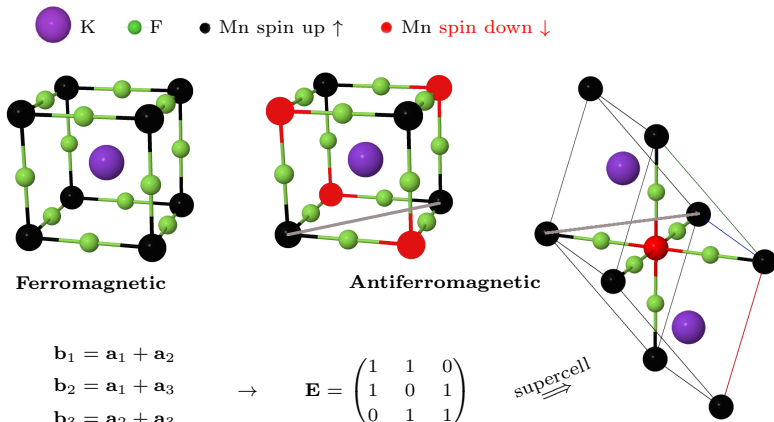
The antiferromagnetic phase of KMnF_3 crystal



The $\mathbf{b}_i$ ($i = 1, 2, 3$) vectors relate the second nearest neighboring Mn atoms to each other (along the face diagonals), whereas next nearest neighbors are no longer equivalent and can carry spin up and down, alternatively, as is shown in the figure above.

As the cell is doubled, the number of atoms per cell doubles too (10 instead of 5). However, the symmetry remains cubic (48 point symmetry operators).

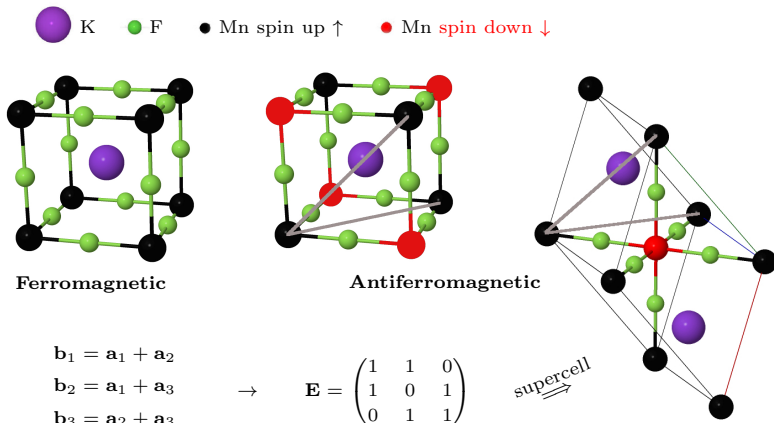
The antiferromagnetic phase of KMnF_3 crystal



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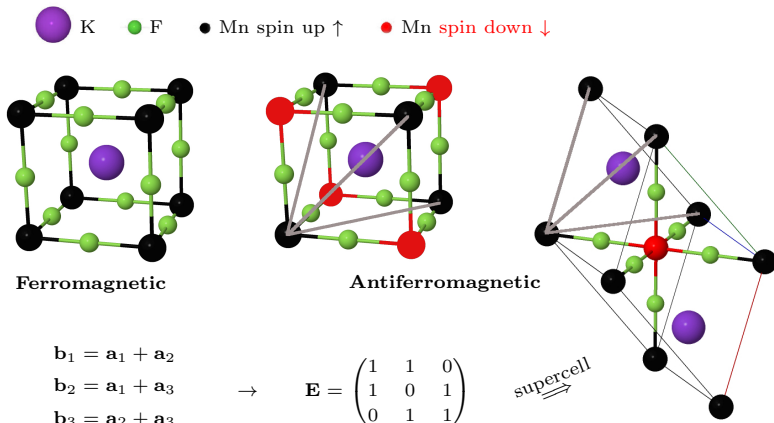
The antiferromagnetic phase of KMnF_3 crystal



The $\mathbf{b}_i$ ($i = 1, 2, 3$) vectors relate the second nearest neighboring Mn atoms to each other (along the face diagonals), whereas next nearest neighbors are no longer equivalent and can carry spin up and down, alternatively, as is shown in the figure above.

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As the cell is doubled, the number of atoms per cell doubles too (10 instead of 5). However, the symmetry remains cubic (48 point symmetry operators).

The antiferromagnetic phase of KMnF_3 crystal - SUPERCELL keyword

The easiest way to obtain the antiferromagnetic phase of KMnF_3 crystal is to use the **SUPERCELL** option to define the geometry of the appropriate double volume cell.

$$\begin{aligned}\mathbf{b}_1 &= e_{11}\mathbf{a}_1 + e_{12}\mathbf{a}_2 + e_{13}\mathbf{a}_3 \\ \mathbf{b}_2 &= e_{21}\mathbf{a}_1 + e_{22}\mathbf{a}_2 + e_{23}\mathbf{a}_3 \\ \mathbf{b}_3 &= e_{31}\mathbf{a}_1 + e_{32}\mathbf{a}_2 + e_{33}\mathbf{a}_3\end{aligned}\quad \mathbf{E} = \begin{pmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{pmatrix}$$

The input cards to be added in the geometry input section are the following.

[in the first block : geometry setup block]	
SUPERCELL	Keyword to create a supercell
1 1 0	(e_{11}, e_{12}, e_{13}) components of the expansion matrix
1 0 1	(e_{21}, e_{22}, e_{23}) components of the expansion matrix
0 1 1	(e_{31}, e_{32}, e_{33}) components of the expansion matrix
END	End of the first block input section

Using the expansion matrix defined in the previous example, the direct lattice vectors of the primitive cell become

$$\mathbf{E} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \rightarrow \begin{aligned}\mathbf{b}_1 &= \mathbf{a}_1 + \mathbf{a}_2 \\ \mathbf{b}_2 &= \mathbf{a}_1 + \mathbf{a}_3 \\ \mathbf{b}_3 &= \mathbf{a}_2 + \mathbf{a}_3\end{aligned}$$

The antiferromagnetic phase of KMnF_3 crystal - SPINLOCK keyword

Exercise 4. SPINLOCK keyword to study the antiferromagnetic phase of KMnF_3 crystal.

Using the input file $\text{KMnF}_3.\text{d12}$ as a starting point, follow the steps listed below.

- Run CRYSTAL for the antiferromagnetic phase of KMnF_3 specifying
 1. the keyword SUPERCELL to set the correct geometric structure of the unit cell
 2. the keyword SPINLOCK to force the ground state solution toward the condition
$$S_z^{\text{cell}} = 0 \quad \rightarrow \quad n_\alpha - n_\beta = 0 \quad \text{for the first 50 iterations of the SCF cycle}$$
- Analyze the output file to check if the antiferromagnetic solution has been obtained.

Pay attention: the number of atoms and the dimension of the unit cell constructed for the antiferromagnetic phase is different from the case of the ferromagnetic phase, so that the SHRINK factor has to be conveniently modified.

The antiferromagnetic phase of KMnF_3 crystal - SPINLOCK keyword

Exercise 4. SPINLOCK keyword to study the antiferromagnetic phase of KMnF_3 crystal.

Using the input file $\text{KMnF}_3.\text{d12}$ as a starting point, follow the steps listed below.

- Run CRYSTAL for the antiferromagnetic phase of KMnF_3 specifying
 1. the keyword SUPERCELL to set the correct geometric structure of the unit cell
 2. the keyword SPINLOCK to force the ground state solution toward the condition
$$S_z^{\text{cell}} = 0 \quad \rightarrow \quad n_\alpha - n_\beta = 0 \quad \text{for the first 50 iterations of the SCF cycle}$$
- Analyze the output file to check if the antiferromagnetic solution has been obtained.

Pay attention: the number of atoms and the dimension of the unit cell constructed for the antiferromagnetic phase is different from the case of the ferromagnetic phase, so that the SHRINK factor has to be conveniently modified.

Solution.

The summed spin density of the unit cell is correctly equal to zero, but the spin density is zero everywhere in the cell (diamagnetic or closed shell system) and not localized on the Mn ions. Therefore, the antiferromagnetic solution is not obtained.

The antiferromagnetic phase of KMnF_3 crystal - SPINLOCK keyword

In the antiferromagnetic case, the SPINLOCK directive will specify that

$$S_z^{\text{cell}} = 0 \quad \rightarrow \quad n_\alpha - n_\beta = 0 \quad (1)$$

Ferromagnetic	Antiferromagnetic	third block : single particle Hamiltonian and SCF control
SPINLOCK	SPINLOCK	Keyword to lock the spin difference $n_\alpha - n_\beta$
5 30	0 30	$n_\alpha - n_\beta$ and number of SCF cycles with $n_\alpha - n_\beta$ fixed
END	END	End of the third block input section

However, in this case the SPINLOCK keyword is not sufficient to obtain an antiferromagnetic solution

Actually, the condition (1) can be satisfied in two different ways:

- the spin density is zero everywhere in the cell (diamagnetic or closed shell system)
- the spin density is not zero, but there are as many α as β spin electrons (any number)

With this indeterminacy, it is usually difficult to converge the self-consistent cycle to any definite state with CRYSTAL and, even if one succeeds in converging to some solution, it will generally not be any of the possible antiferromagnetic phases, but most likely some metallic state with much higher total energy per cell.

The reason for this nasty behavior is that the **initial Fock (Kohn-Sham) density matrix defined in this case is too far from the final self-consistent solution** $\rightarrow$ $\rightarrow$ it is always helpful (in many cases mandatory) to drive the system to the required solution through the choice of a more **reasonable initial guess**

The antiferromagnetic phase of KMnF_3 crystal - ATOMSPIN keyword

Solution a. Start the self-consistent cycle from the isolated atoms solution and use the **ATOMSPIN** directive.

Initial guess : isolated atoms + ATOMSPIN keyword for spin inversion

The keyword ATOMSPIN assigns the unpaired electrons of one or more atoms in the initial guess to the desired spin component, as described in the following table.

[in the third block : single particle Hamiltonian and SCF control block]	
ATOMSPIN	Keyword to set the atomic spin
2	number of atoms to attribute a spin
1 +1 2 -1	atom labels (output file list order) and spin (1, 0, -1)
END	End of the third block input section

In this way CRYSTAL defines open shell configurations for two atoms in the initial guess:

- the unpaired electrons of the first atom in the output list have spin α (+1)
- the unpaired electrons of the second atom in the output list have spin β (-1)

If not specified, all equivalent atoms would be attributed the same spin (α or β).

In the case of the KMnF_3 supercell previously created $\rightarrow$ one of the two Mn ions in the cell has spin up (the first in the output list), while the other Mn ion (the second in the output list) has spin down.

The antiferromagnetic phase of KMnF_3 crystal - ATOMSPIN keyword

Exercise 5. ATOMSPIN keyword to study the antiferromagnetic phase of KMnF_3 crystal.

Using the input file of Exercise 4 as a starting point, follow the steps listed below.

- Run CRYSTAL for the antiferromagnetic phase of KMnF_3 specifying
 1. the keyword SUPERCELL to set the correct geometric structure of the unit cell
 2. the keyword SPINLOCK to force the ground state solution toward the condition
$$S_z^{\text{cell}} = 0 \quad \rightarrow \quad n_\alpha - n_\beta = 0 \quad \text{for the first 30 iterations of the SCF cycle}$$
 3. the keyword ATOMSPIN to assign the unpaired electrons of the two Mn ions in the initial guess to the desired spin component so that the antiferromagnetic phase can be obtained
- Analyze the output file to check if the antiferromagnetic solution has been obtained.

The antiferromagnetic phase of KMnF_3 crystal - ATOMSPIN keyword

Exercise 5. ATOMSPIN keyword to study the antiferromagnetic phase of KMnF_3 crystal.

Using the input file of Exercise 4 as a starting point, follow the steps listed below.

- Run CRYSTAL for the antiferromagnetic phase of KMnF_3 specifying
 1. the keyword SUPERCELL to set the correct geometric structure of the unit cell
 2. the keyword SPINLOCK to force the ground state solution toward the condition
$$S_z^{\text{cell}} = 0 \quad \rightarrow \quad n_\alpha - n_\beta = 0 \quad \text{for the first 30 iterations of the SCF cycle}$$
 3. the keyword ATOMSPIN to assign the unpaired electrons of the two Mn ions in the initial guess to the desired spin component so that the antiferromagnetic phase can be obtained
- Analyze the output file to check if the antiferromagnetic solution has been obtained.

Solution.

The self-consistent process converges after few iterations to an antiferromagnetic solution. The summed spin density of the unit cell is equal to zero as a result of the presence of five unpaired electrons with spin up $\uparrow$ localized on the first Mn atom and five unpaired electrons with spin down $\downarrow$ localized on the second Mn atom.

The antiferromagnetic phase of KMnF_3 crystal - SPINEDIT keyword

Solution b. Start the self-consistent cycle from the corresponding ferromagnetic solution and use the **SPINEDIT** directive to apply spin inversion to the unpaired electrons.

Initial guess : ferromagnetic solution + SPINEDIT keyword for spin inversion

Note : The ferromagnetic solution must have the **same geometry** as that of the antiferromagnetic phase (same supercell and number of symmetry operators).

How to start the self-consistent cycle from the ferromagnetic solution?

1. Run a self-consistent calculation to obtain the ferromagnetic solution using the same supercell geometry as for the antiferromagnetic case → **wavefunction in fort.9 file**
2. Run a second self-consistent calculation specifying the **GUESSP** directive, so that the ferromagnetic solution of the previous run is used as starting point for the new self-consistent calculation (**wavefunction of the previous run is read from fort.20 file**)

In this way, the self-consistent calculation for the antiferromagnetic case is started from the density matrix of the corresponding ferromagnetic phase and the SPINEDIT keyword is used to apply spin inversion to the unpaired electrons of one Mn ion in the supercell.

[in the third block : single particle Hamiltonian and SCF control block]

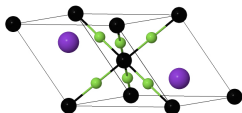
GUESSP	Keyword to read the density matrix from a previous run
SPINEDIT	Keyword to edit the spin density matrix used as SCF guess
1	number of atoms for which spin must be reversed
2	atom labels as given in the output file list
END	End of the third block input section

Use of a density matrix guess for the SCF cycle - GUESSP keyword

- Run a self-consistent calculation of the ferromagnetic phase with the same supercell geometry as the antiferromagnetic phase

Input file name :

input_ferro.d12



Ground state
electronic
density matrix
written in file
fort.9
(scratch folder)

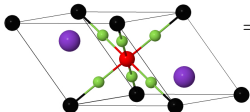


The script runCRY23
renames fort.9 file as
input_ferro.f9

- Run a self-consistent calculation of the antiferromagnetic phase restarting the density matrix from the previous ferromagnetic phase and modifying the spin state

Input file name :

input_antiferro.d12
GUESSP



The script runCRY23
takes the previous file
input_ferro.f9
and it renames it as
fort.20
(scratch folder)



Ground state
density matrix
of ferromagnetic
calculation
read from file
fort.20
as input guess
for the SCF cycle

The antiferromagnetic phase of KMnF_3 crystal - SPINEDIT keyword

Exercise 6. SPINEDIT keyword to study the antiferromagnetic phase of KMnF_3 crystal.

Using the input file of Exercise 1 as a starting point, follow the steps listed below.

- Run CRYSTAL for the ferromagnetic solution phase of KMnF_3 using
 1. the keyword SUPERCELL to set the same geometry of the antiferromagnetic phase

Using the input file of Exercise 4 as a starting point, follow the steps listed below.

- Run CRYSTAL for the antiferromagnetic phase of KMnF_3 specifying
 1. the keyword SUPERCELL to set the correct geometric structure of the unit cell
 2. the keyword SPINLOCK to force the ground state solution toward the condition
$$S_z^{\text{cell}} = 0 \quad \rightarrow \quad n_\alpha - n_\beta = 0 \quad \text{for the first 30 iterations of the SCF cycle}$$
 3. the keyword GUESSP to start the SCF cycle from the previous fort.9 wavefunction
 4. the keyword SPINEDIT to edit the spin density matrix used as SCF guess
- Analyze the output file to check if the antiferromagnetic solution has been obtained.

Exercise 7. Energy difference between ferromagnetic and antiferromagnetic phases.

- Use the results of the Exercise 1 with the same SUPERCELL of Exercise 5 $\rightarrow$ to find the ground state energy E_{FM} of the ferromagnetic phase of KMnF_3 with the same supercell geometry used to study the antiferromagnetic phase
- Use the results of the Exercise 5 or Exercise 6 $\rightarrow$ to find the ground state energy E_{AFM} of the antiferromagnetic phase of KMnF_3
- Use the previous two results to calculate the energy difference (relative stability) between the ferromagnetic and the antiferromagnetic phases for KMnF_3 as

$$\Delta E = |E_{\text{FM}} - E_{\text{AFM}}|$$

Which phase is the most stable one?

Exercise 7. Energy difference between ferromagnetic and antiferromagnetic phases.

- Use the results of the Exercise 1 with the same SUPERCELL of Exercise 5 $\rightarrow$ to find the ground state energy E_{FM} of the ferromagnetic phase of KMnF_3 with the same supercell geometry used to study the antiferromagnetic phase
- Use the results of the Exercise 5 or Exercise 6 $\rightarrow$ to find the ground state energy E_{AFM} of the antiferromagnetic phase of KMnF_3
- Use the previous two results to calculate the energy difference (relative stability) between the ferromagnetic and the antiferromagnetic phases for KMnF_3 as

$$\Delta E = |E_{\text{FM}} - E_{\text{AFM}}|$$

Which phase is the most stable one?

Solution.

From the results of Exercise 1 $\rightarrow E_{\text{FM}} = -4095.2868338 \text{ Ha}$

From the results of Exercise 5 $\rightarrow E_{\text{AFM}} = -4095.2874226 \text{ Ha}$

$$\Delta E = |E_{\text{FM}} - E_{\text{AFM}}| = 5.888 \cdot 10^{-4} \text{ Ha} \quad \text{with} \quad E_{\text{FM}} > E_{\text{AFM}}$$

Therefore, the antiferromagnetic phase is more stable than the ferromagnetic phase

Accuracy of the magnetic phases ground state energy calculation

Since magnetic phase transitions involve quite small energy variations ΔE , of the order of 10^{-4} Hartree, special care must be paid in the determination of the total energy for the two phases. If the ground state total energy of the two phase are accurate enough, then meaningful energy differences can be computed.

In order to **check the accuracy of the total energy for the two phases** the following tests can be done.

- **Does the use of supercells affect ΔE determination?**

The ground state total energy of the ferromagnetic phase obtained in Exercise 1 is

$$E_{\text{FM}}^u = -2047.6434177 \text{ Ha} \quad (\text{unit cell} \rightarrow 1 \text{ Mn ion i.e. 5 unpaired electrons})$$

The ground state total energy of the ferromagnetic phase with the same supercell geometry of that implied to study the antiferromagnetic phase in Exercise 5 is

$$E_{\text{FM}}^s = -4095.2868338 \text{ Ha} \quad (\text{supercell} \rightarrow 2 \text{ Mn ion i.e. 10 unpaired electrons})$$

Therefore, the supercell energy is almost exactly twice the unit cell energy

$$E_{\text{FM}}^s - 2E_{\text{FM}}^u = 1.6 \cdot 10^{-6} \ll 5.888 \cdot 10^{-4} = \Delta E$$

This is one of the method to evaluate the numerical accuracy of a periodic code: compute the total energy of the unit cell of a system and then the total energy of supercells of the same system with increasing size. The total energy should obviously scale linearly with the volume or the number of atoms.

Accuracy of the magnetic phases ground state energy calculation

Since magnetic phase transitions involve quite small energy variations ΔE , of the order of 10^{-4} Hartree, special care must be paid in the determination of the total energy for the two phases. If the ground state total energy of the two phase are accurate enough, then meaningful energy differences can be computed.

In order to **check the accuracy of the total energy for the two phases** the following tests can be done.

- **Experimental or calculated equilibrium geometry - which is better?**

The small value of the energy difference ΔE between the ferromagnetic and the antiferromagnetic phases certainly leads to exclude a priori that any significant geometry variation may take place during the ferromagnetic - antiferromagnetic transition, so that both phases can be described as having essentially the same geometry.

On the other hand, as ΔE decreases with the Mn-Mn distance very rapidly, small errors in the determination of equilibrium geometry (typically of the order of 2% within the Hartree-Fock approximation) may affect ΔE with large errors.

For these reasons, working at the experimental geometry is certainly to be preferred with respect to a structural optimization strategy.

The magnetic exchange coupling constant

The calculation of the magnetic exchange coupling constant J is particularly important for comparison with experiment.

To relate the exchange coupling constant J with the ground state energy of the three-dimensional periodic crystalline systems, the starting point can be the quantum Heisenberg model, described by the Hamiltonian

Heisenberg Hamiltonian :
$$\mathcal{H} = -J \sum_{i=1}^m \mathbf{S}_i \cdot \mathbf{S}_{i+1} \quad (2)$$

where the summation goes over each nearest neighbor pair of the m magnetic sites on a lattice, taken once, and $\mathbf{S}_i$ are spin vectors, usually represented in terms of their Hermitian Cartesian component operators. In order to further simplify the model, it can be assumed that the spin-spin interaction takes place through only one magnetic component (along the z direction), that is

$$\mathbf{S}_i = (0, 0, S_i^z) \quad \forall i = 1, \dots, m \quad \rightarrow \quad \mathcal{H} = -J \sum_{i=1}^m S_i^z S_{i+1}^z \quad (3)$$

Although, in general, this is not the case, the application of this last model is much easier than the use of Heisenberg Hamiltonian and more compatible with the monodeterminantal solution, which corresponds to non pure spin state for general open-shell systems. In addition, it has been shown[3] that equation (3) is obtained from the Heisenberg Hamiltonian when the mean field approximation is applied.

[3] Yen-Sheng Su, T.A. Kaplan, S.D. Mahanti and J.F. Harrison, Phys. Rev. B59, 10521 (1999)

The magnetic exchange coupling constant of KMnF_3 crystal

$$\mathcal{H} = -J \sum_{i=1}^m S_i^z S_{i+1}^z \quad (4)$$

Equation (4) can be further simplified to compute the exchange coupling constant of KMnF_3 crystalline system using the following approximations

- (i) since our model is periodic, the spin density at all atoms other than Mn can be neglected, and the summation in the equation can be limited to the Mn ions of one unit cell, that are symmetric (or spin-antisymmetric)
- (ii) considering that the super-exchange interaction is short ranged, as a first approximation, coupling with only next nearest neighbors Mn ions needs to be included

Computing the ground state eigenvalue of the spin Hamiltonian (4) in both the ferromagnetic ($J > 0$) and antiferromagnetic ($J < 0$) phases, and subtracting these eigenvalues, the following relation between the energy difference ΔE of the two phases and J for the ferromagnetic-antiferromagnetic transition can be obtained

$$\Delta E = |E_{\text{FM}} - E_{\text{AFM}}| = J N Z (S_{\text{Mn}}^z)^2 \quad (5)$$

where N is the number of symmetric Mn atoms in the (super)cell, Z is the number of the next nearest (Mn) neighbors to each of the N Mn with opposite spin and S_{Mn}^z is the Mn spin component along the z direction.

The magnetic exchange coupling constant of KMnF_3 crystal

Exercise 8. Calculation of the magnetic exchange coupling constant of KMnF_3 crystal.

Use the result of Exercise 7 to calculate the magnetic exchange coupling constant J for the ferromagnetic-antiferromagnetic transition using the formula derived by a simplified version of the Heisenberg model

$$J = \frac{k \Delta E}{N Z (S_{\text{Mn}}^z)^2} \quad \text{where} \quad \Delta E = |E_{\text{FM}} - E_{\text{AFM}}|$$

N : number of symmetric Mn atoms in the (super)cell

Z : number of the next nearest Mn neighbors to each of the N Mn with opposite spin

S_{Mn}^z : spin component of each Mn atom along the z direction

k : conversion factor from Hartree to Kelvin that can be obtained as follows

$$k = \frac{\text{conversion factor Ha} \rightarrow \text{J}}{\text{Boltzmann constant}} = \frac{4.359748 \cdot 10^{-18} \text{ J}}{1.3806558 \cdot 10^{-23} \text{ J/K}} = 3.1577 \cdot 10^5 \text{ K}$$

Compare the result with the one obtained in Ref. [1] and with the experimental values reported in the same reference.

[1] R. Dovesi, F. Freyria Fava, C. Roetti and V.R. Saunders, Faraday Discuss. 106, 173 (1997)
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